

Impedance Modeling of Solid-State Electrolytes: Influence of the Contacted Space Charge Layer

Yao Liu, Yang Bai, Wolfram Jaegermann, René Hausbrand,* and Bai-Xiang Xu*

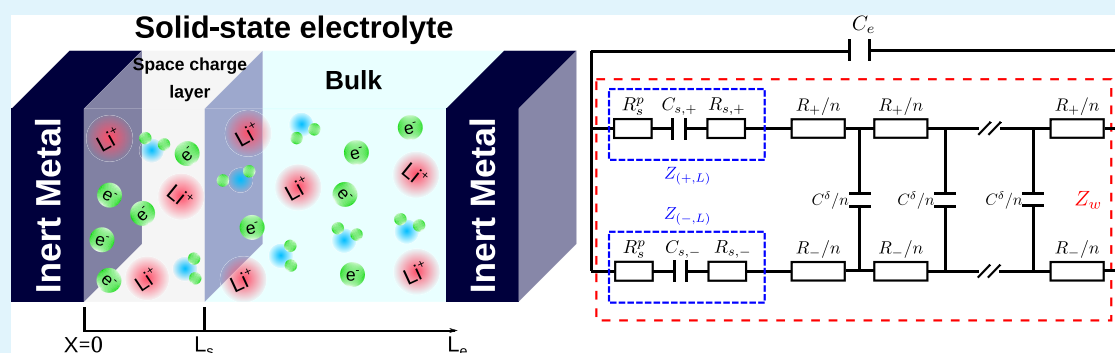
Cite This: *ACS Appl. Mater. Interfaces* 2021, 13, 5895–5906

Read Online

ACCESS |

Metrics & More

Article Recommendations



ABSTRACT: Understanding the interfacial impedance between the solid electrolyte and the electrode is a critical issue for the design of solid-state batteries. We propose a new equivalent circuit model that treats the interface not only as a capacitor but also includes the space charge layer resistance and the resultant polarization resistance. Moreover, the elements of the circuit model are quantified by the physical quantities based on the recently proposed modified Planck–Nernst–Poisson (MPNP) model, which includes the effect of the unoccupied regular lattice sites (vacancies) in the electro-diffusion problem and takes both the ion and electron contributions into the account. We provide a new analytical solution for the space charge layer capacitance. Comparative numerical results demonstrate that our proposed model with additional polarization resistance can explain well the real impedance tail at the low-frequency region, for which the pure capacitor interface model fails. The model is verified against the experimental impedance spectra of LiPON.

KEYWORDS: interfacial impedance, MPNP model, equivalent circuit model, space charge layer capacitance, polarization effect

1. INTRODUCTION

Rechargeable lithium-ion batteries are widely used in power-intensive portable devices, electric vehicles, and renewable energy storage because of their high energy and power densities.^{1–4} However, the organic solvent electrolyte in the classical lithium-ion battery system poses a safety issue and manufacturing difficulties.⁵ More attention has been recently given to the development of lithium-ion solid-state electrolyte batteries due to the increased safety and the operational range.^{6,7} However, the solid-state electrolyte has a relatively high interfacial impedance, which is the bottleneck for the development of all-solid-state batteries.⁸ With a potential perturbation, charges accumulate in the space charge layer and lead to a drastic electrostatic potential drop. The understanding of the space charge layer impedance is very limited.^{9–13} Luntz¹⁰ pointed out that the principal obstacle to developing successful solid-state batteries is not related to a maximum of the ionic conductivity but in minimizing the interfacial impedance. Nevertheless, Klerk⁹ mentioned that the impedance for lithium-ion transport through the space charge layer is negligible, except when a

complete lithium-ion depletion layer forms in the solid-state electrolytes. Therefore, the influence of the space charge layer in the solid-state electrolytes is still an open question for researchers. To tackle this critical issue, electrochemical impedance spectroscopy (EIS) is a well-established technique for the characterization and measurement of solid-state batteries impedance. However, the EIS is only able to give the total impedance of solid-state batteries and it is hard to distinguish the contribution of each part, e.g., the bulk resistance and the space charge layer capacitance. The equivalent circuit model is thus introduced as a method to interpret the electrochemical impedance spectroscopy. By fitting the experimental results with the estimated impedance spectroscopy, partial elements in

Received: December 29, 2020

Accepted: January 12, 2021

Published: January 22, 2021



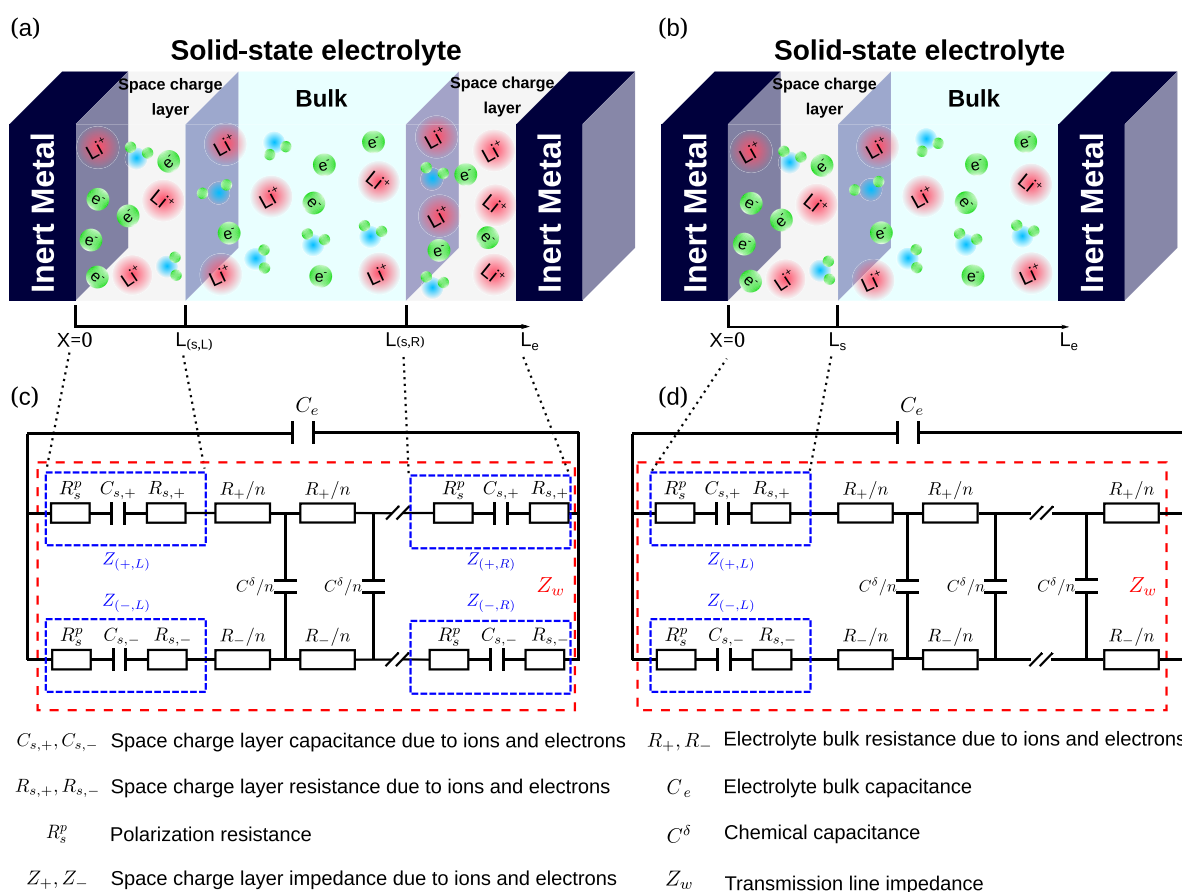


Figure 1. (a) Schematic of a metal/electrolyte/metal cell with the space charge layers at the interface and (c) the proposed equivalent circuit model. (b) Schematic of a cell with an infinite solid-state electrolyte assumption ($L_s \ll L_e$) and (d) the corresponding equivalent circuit model. It should be noticed that the infinite electrolyte assumption and the circuit model (d) are adopted in our work.

the equivalent circuit model can be quantified and utilized to analyze the overall battery impedance.¹⁴ The equivalent circuit model is widely applied, but it should be noticed that different circuit models can produce the same impedance results.¹⁵

Elements in the equivalent circuit models have intrinsic correlations with material properties such as diffusivity of charge carriers. The quantification of the elements from the material properties have not been addressed sufficiently, for which reliable physical models are indispensable. To figure out the intrinsic relations, researchers have to utilize electrochemical models to calculate the charge and the electrostatic potential distribution inside the batteries, especially in the space charge layer.^{16–20} The standard Planck–Nernst–Poisson (PNP) model²¹ is commonly applied for all-solid-state batteries to predict the concentration, and thus the corresponding equivalent circuit model has been developed for impedance simulation based on the standard PNP framework.^{22–24} Moreover, the charge density and the thickness of the space charge layer have a significant impact on the interface capacitor and resistance.^{25,26} However, as mentioned in our previous work,²⁷ the unoccupied regular lattice sites (vacancies) in the solid-state electrolyte is not taken into consideration in the standard PNP model. The calculated concentration in the space charge layer will exceed the maximum lattice sites afforded by real materials at a high potential perturbation. Thus, the space charge layer capacitance and resistance derived from the standard PNP model are questionable and have to be discussed seriously. Furthermore, the space charge layer is simply regarded

as an ideal capacitor in some investigations^{20,28} and this assumption does not correlate well with the experimental observations at low frequencies.

To solve these problems and accurately quantify the space charge layer impedance, we apply the modified Planck–Nernst–Poisson (MPNP) model^{27,29} in this work to derive a new equivalent circuit model. Within this MPNP model, the vacancies in the solid-state electrolyte³⁰ are taken into consideration and the charge density in the space charge layer is precisely calculated. From the MPNP model, we first derive a more accurate analytical solution for the space charge layer capacitance. Moreover, the polarization effect at the interface, which has been observed in the experimental results,^{31,32} has been hardly discussed in the impedance simulation. Shown in the literature,^{33,34} the charge density and the interfacial impedance in the space charge layer are frequency-dependent. Thus, we take a further step and introduce the polarization impedance in the space charge layer for the first time to consider this factor. The corresponding equivalent circuit model is presented, and the physical elements are well interpreted by the material properties. Moreover, this equivalent circuit model is a general model for the solid-state electrolyte and provides a perspective to identify the origin of high interfacial resistances and to design outstanding solid-state batteries. The structure of the paper is organized as follows: in the impedance modeling section, the impedance calculation methodology and the set of equations are discussed. The results section describes the corresponding physical results such as the analytical diffuse layer

capacitance, the polarization resistance, and the chemical capacitance. Here, we introduce the equivalent circuit model and the comparison of numerical and experimental results. We conclude with a summary of our work and an outlook in Section 4.

2. IMPEDANCE MODELING

We study a metal/electrolyte/metal cell consisting of a solid-state electrolyte and a pair of electrodes. Figure 1a and Figure 1b are the schematics of a real situation and a case with the infinite electrolyte assumption, respectively. This cell configuration is extensively applied to measure the solid-state electrolyte impedance spectroscopy individually. To verify our model and numerical results with the experimental data^{31,32} in the later section, the amorphous material LiPON is considered as the solid-state electrolyte with a thickness L_e and the electrochemically inert metal Pt is chosen as the electrode. With an external disturbance, charges accumulate at the metal/electrolyte interface and form a space charge layer as shown in Figure 1b,d.^{35–37} The thickness of the space charge layer is denoted by L_s and is of the order of 1–10 nm.³⁸

2.1. Equivalent Circuit Model. Figure 1c,d illustrates the proposed equivalent circuit model for the metal/electrolyte/metal cell configuration, each element of which will be directly correlated to the solid-state electrolyte material properties in the following sections. According to the investigations,^{32,39,40} electrons can move with a relative low velocity in the solid-state electrolyte at room temperature, thus, the contributions of the electrons are also considered in the equivalent circuit model to maintain the generality. The elements related to the ions are indicated by the subscript +, while the electrons are denoted by –. In the following context, the subscript i represent either + or –. The quantities R_i and C_i refer to the electrolyte bulk resistance and dielectric capacitance contributed by the ions and electrons, respectively. C^δ is the electrolyte chemical capacitance, and n represents the number of transmission line elements and is defined as $n = L_e/x$, where x is the element thickness.^{41,42}

The charge accumulation at the metal/electrolyte interface contributes to the space charge layer impedance Z_ν and thus the impedance is proportional to the charge density. In order to focus on the interface layer, we consider only the left space charge layer impedance $Z_{(i, L)}$, while the right-hand space charge layer impedance $Z_{(i, R)}$ is ignored,^{17,20,43} as it is shown in Figure 1d. The simplification corresponds to the case of an infinite solid-state electrolyte thickness and is reasonable when the space charge layer is much smaller than the electrolyte ($L_s \ll L_e$). Note that a similar impedance analysis can be applied to the right-hand electrolyte interface if it is included. From a thermodynamic point of view, the chemical capacitor C_i^δ is purely non-electrostatic and is defined as second derivative of the Gibbs free energy with respect to the number of species. Thus, C_i^δ is given by the changes of the component chemical potential μ_i due to the concentration c_i variations.^{22,41} In the solid-state electrolyte, the stoichiometry effect varies significantly when the density of vacancies are included. Moreover, the C_i^δ is proportional to the material volume and can be very large compared to the electrolyte bulk capacitance C_i . The chemical capacitance C^δ line is omitted at high frequencies, thus, the transmission line Z_w is simplified by the ionic line R_+ in parallel with the electronic R_- line in Figure 1d. Nevertheless, the chemical capacitance C^δ cannot be ignored at low frequencies in the equivalent circuit model. In some investigations, the space charge layer impedance Z_i is always assumed as an ideal capacitor^{20,31,44} or can be

expressed as the space charge layer capacitance $C_{s, i}$ in parallel with the space charge layer resistance $R_{s, i}$.^{33,45,46} However, MacDonald³⁴ pointed out that the space charge layer impedance Z_i follows the relation $Z_i = (j\omega C_{s, i})^{-1} + R_{s, i}$. Note that the space charge layer impedance Z_i is proportional to the charge density; thus, the capacitor $C_{s, i}$ and the resistance $R_{s, i}$ are frequency-dependent. The results can only be resolved numerically and fails to correlate with material properties; thus, researchers are prone to calculate the values at the equilibrium state, i.e., the frequency equals zero and simplify the explanation. This simplification ignores the polarization effect and frequency perturbation; thus, we introduce the polarization resistance and hope to solve this problem. As shown in Figure 1d, the space charge layer impedance Z_i consists of the space charge layer capacitance $C_{s, i}$, the space charge layer resistance $R_{s, i}$ and the space charge layer polarization resistance R_s^p , i.e., $Z_i = (j\omega C_{s, i})^{-1} + R_{s, i} + R_s^p$.

2.2. Methodology of Impedance Simulation. In electrochemical impedance spectroscopy measurement, the widely adopted method is the application of a small perturbation potential or current to an electrochemical system and measuring the corresponding response. The response is the ratio of the perturbation potential to the current and is described as an impedance Z .^{47,48} In this work, the metal/electrolyte/metal (Pt/LiPON/Pt) cell is operated with a pulse electric potential condition and the formula is given by

$$\Phi = \Phi_{dc} + \Phi_{ac}(e^{j\omega t}) \quad (1)$$

where Φ_{dc} is the potential with a direct current condition, Φ_{ac} is the alternating current potential, j is the imaginary number and equals to $\sqrt{-1}$, and ω is the frequency of the applied alternating current potential. According to the previous investigations,^{47,48} the corresponding current also has the same type formula and is depicted by

$$I = I_{dc} + I_{ac}[e^{j(\omega - \omega_0)t}] \quad (2)$$

Here, I_{dc} corresponds to the steady-state current and is time independent, I_{ac} is the perturbation current, ω_0 is the phase angle difference between the imposed electric potential and the measurement current in eqs 1 and 2. Note that the electrochemical impedance Z is defined as¹⁷

$$Z = \frac{\Phi_{ac}}{I_{ac}} = Z_{re}(\omega) + j \cdot Z_{im}(\omega) \quad (3)$$

where Z_{re} and Z_{im} are the real and imaginary parts of the complex impedance, respectively.

2.3. Numerical Model. As shown in Figure 1b, the charge distribution in the solid-state electrolyte is related to the equivalent circuit elements and has a large impact on the impedance spectroscopy. Thus, the precise prediction of the charge accumulation in the space charge layer is particularly crucial and deserves a careful discussion. As presented in our previous work,²⁷ the modified Planck–Nernst–Poisson (MPNP) model shows an apparent advantage over the standard PNP model. The limited amount of unoccupied regular lattice sites (vacancies) in the solid-state electrolyte are considered within the MPNP model and equations are given by

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot J_i \quad J_i = -\frac{D_i}{1 - \tilde{c}_i} \nabla c_i - z_i \frac{FD_i}{RT} c_i \nabla \Phi \quad (4)$$

$$\nabla^2 \Phi = -\frac{F}{\epsilon_0 \epsilon_i} (z_+ c_+ + z_- c_- - z \rho) \quad (5)$$

Hereby, D_i is the diffusivity and J_i is the flux of species i , c_i is the mobile concentration of i and $\tilde{c}_i$ is the corresponding normalized concentration as shown in eq 8, F is the Faraday constant, R is the gas constant, and T is the temperature. Moreover, the electronic conductivity is considered and the migration of electrons is expressed by eq 4 with the valence $z_i = -1$. The symbol ρ denotes the immobile charge, z_i is the valence of the species i and z equals to $z = |z_i|$, Φ is the electrostatic potential, and ϵ_i and ϵ_0 are the relative and vacuum permittivities of the solid-state electrolyte, respectively.

The total electrical current I in the solid-state electrolyte consists of the faradaic current I_f and the Maxwell displacement current I_d , which is expressed by $I = I_f + I_d$. The faradaic current is attributed to the species flux in the solid-state electrolyte and is given by

$$I_f = FA \sum_i z_i J_i \quad (6)$$

where A is the geometrical surface area of the cell. The Maxwell displacement current results from the frequency of electric field strength and is depicted by

$$I_d = -A \epsilon_0 \epsilon_i \frac{\partial(\nabla \Phi)}{\partial t} \quad (7)$$

To simplify the discussion and ensure the impedance simulation more mathematically tractable, several assumptions and boundary conditions follow the literature and are applied.^{17,20} (1) The metal/electrolyte/metal cell is considered as one-dimensional domain and operated with an alternating current (ac) potential perturbation in the frequency range. (2) The electrostatic potential drop and the species migration in the inert metal are ignored; thus, the boundary conditions for the solid-state electrolyte are $\Phi_L = \Phi_{dc} + \Phi_{ac}(e^{j\omega t})$ and $\Phi_R = 0$. (3) Lithium ions and electrons in the solid-state electrolyte are completely blocked at the left-hand side. Therefore, the boundary conditions of the species concentrations are $J_{(+, L)} = 0$ and $J_{(-, L)} = 0$. (4) The behaviors at the right-hand side of the solid-state electrolyte are regarded as infinite length behaviors, and no charge separation and space charge layer formation at the right-hand side and the concentrations are $c_{(+, R)} = c_0$ and $c_{(-, R)} = c_0$. (5) No lithium-ion intercalation happens and no contact resistance at the metal/electrolyte interface.

3. RESULTS AND DISCUSSION

The simulated electrochemical impedance spectroscopy and the corresponding material properties quantification are presented in this section. In the analytical subsection, the analytical space charge layer capacitance (C_s) caused by the charge accumulation is derived. Then, the space charge layer impedance compositions, the corresponding equivalent circuit model, and the temperature influences are discussed in particular. Finally, the impedance spectroscopies comparison of the numerical and experimental Pt/LiPON/Pt cell results is shown in the experimental subsection. Numerical simulation has been performed with the finite element method in the MOOSE framework.⁴⁹ All the parameters applied in the impedance simulation are derived from the experimental research of LiPON thin film^{50,51} and are shown in Table 1. To promote the convergence of numerical calculation, normalization of the

Table 1. Solid-State Electrolyte Parameters

parameter	unit	value	description
L_e	nm	100	thickness of the electrolyte
D_+	$\text{m}^2 \text{s}^{-1}$	10^{-14}	diffusivity of lithium ions in the electrolyte
D_+^*	$\text{m}^2 \text{s}^{-1}$	4×10^{-14}	tracer diffusivity of lithium ions in the electrolyte
D_-	$\text{m}^2 \text{s}^{-1}$	10^{-16}	diffusivity of electrons in the electrolyte
D_-^*	$\text{m}^2 \text{s}^{-1}$	4×10^{-16}	tracer diffusivity of electrons in the electrolyte
ϵ_0	F m^{-1}	8.85×10^{-12}	vacuum permittivity
ϵ_i		20	relative permittivity
E_a	eV	0.5	activation energy
σ_0	S m^{-1}	1.59×10^7	pre-exponential factor
σ	S m^{-1}	1.88×10^{-4}	ionic conductivity at 298.15 K
$c_{\max}$	mol m^{-3}	10^4	maximum concentration
c_+	mol m^{-3}	5×10^3	initial mobile lithium ions in the electrolyte
c_-	mol m^{-3}	5×10^3	initial mobile electrons in the electrolyte
F	C mol^{-1}	96,485	Faraday constant
T	K	298.15	temperature
R	$\text{J mol}^{-1} \text{K}^{-1}$	8.314	gas constant
z_+		1	lithium ion valence
z_-		-1	electron valence
A	m^2	4×10^{-6}	geometrical surface area

model is carried out by using the following dimensionless parameters indicated by the tilde hat:

$$\tilde{L} = \frac{L}{L_0} \quad \tilde{t} = \frac{t}{t_0} \quad \tilde{c} = \frac{c}{c_{\max}} \quad \tilde{\Phi} = \frac{F}{RT} \Phi \quad \tilde{D} = \frac{t_0}{L_0^2} D \quad \tilde{\omega} = t_0 \omega \quad (8)$$

3.1. Analytical Capacitance of the Diffuse Layer. The space charge layer capacitance C_s is a critical factor in the interfacial impedance analysis; thus, researchers hope to derive an analytical solution of the capacitance C_s and to correlate with material properties. An analytical solution of the space charge layer capacitance C_s is given in Wang's work²⁰ and provides a good estimation. However, the predicted value shows a discrepancy with the numerical results calculated from the standard or modified PNP models. Thus, an accurate theoretical space charge layer capacitance C_s based on our MPNP model is given in this subsection. In the standard PNP model, the concentration distribution in the space charge layer at the equilibrium state is related to the electrostatic potential by the Boltzmann distribution³⁸ and is given by

$$c_i = c_{i,\infty} \exp\left(\frac{-z_i F \Phi}{RT}\right) \quad (9)$$

where $c_{i,\infty}$ is the bulk concentration of species i in the electrolyte. Due to a limited amount of vacancies in the space charge layer, eq 9 fails to describe the concentration distribution in the MPNP model, and the new formula is expressed by the modified Boltzmann distribution

$$c_i = \frac{c_{i,\infty} e^{-z_i F \Phi / RT}}{1 + \tilde{c}_{i,\infty} (e^{-z_i F \Phi / RT} - 1)} \quad (10)$$

and $\tilde{c}_{i,\infty}$ is the dimensionless concentration. For general consideration, the total concentrations of mobile lithium ions

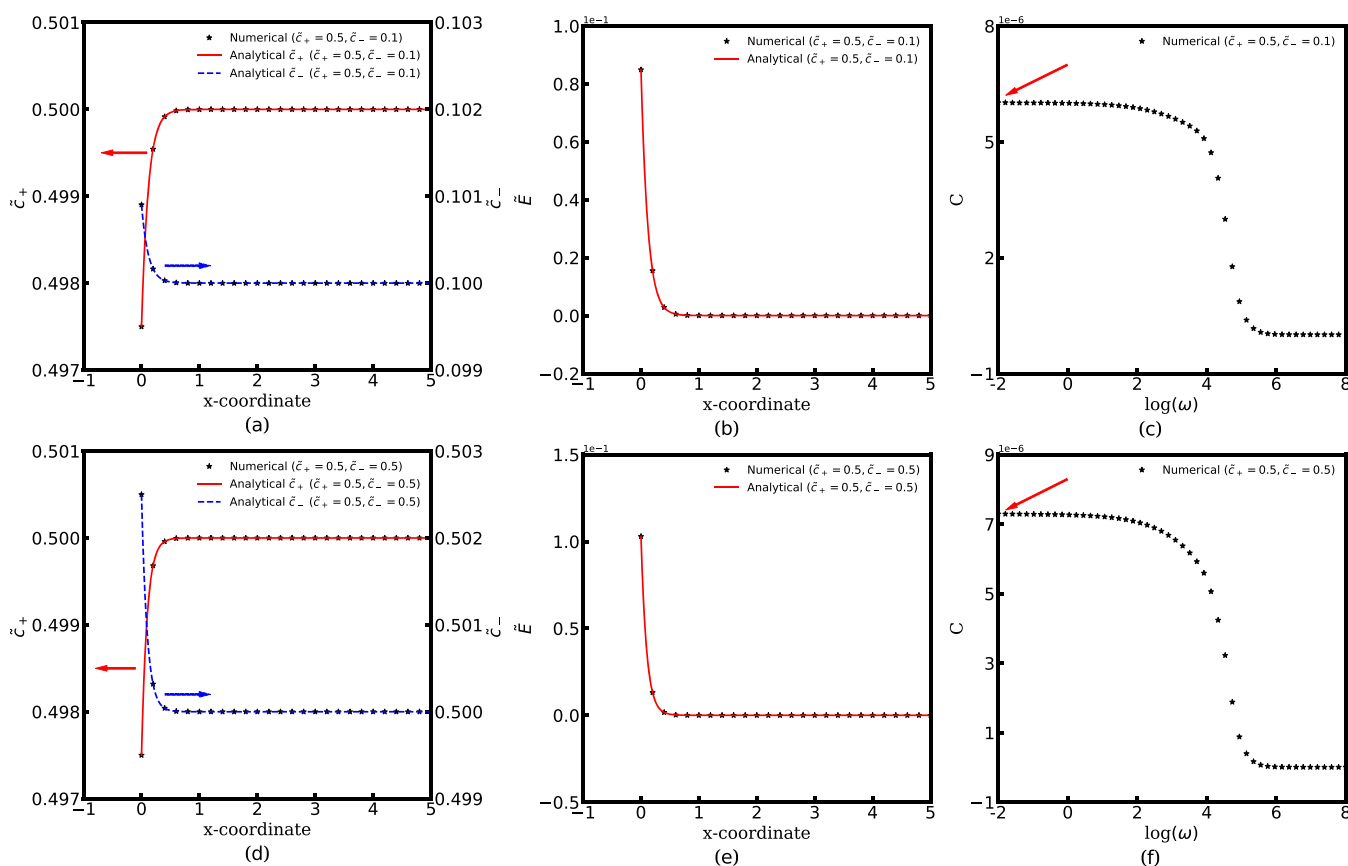


Figure 2. (a, d) Equilibrium concentration of lithium ions $\tilde{c}_+$ and electrons $\tilde{c}_-$, (b, e) electric field $\tilde{E}$ at the metal/electrolyte interface. (c, f) Specific capacitance C as a function of frequencies. The upper figures are $\tilde{c}_+ = 0.5$ and $\tilde{c}_- = 0.1$, and the lower figures denote $\tilde{c}_+ = 0.5$ and $\tilde{c}_- = 0.5$.

and electrons are not necessarily the same in the solid-state electrolyte. The Poisson equation is given in the following with ρ being the charge of the immobile species $\rho = c_+ - c_-$ and z_i the valence, and $z = |z_i|$:

$$\begin{aligned} \nabla^2 \Phi &= -\frac{F}{\epsilon_0 \epsilon_i} (z_+ c_+ + z_- c_- - z\rho) \\ &= -\frac{F}{\epsilon_0 \epsilon_i} \left[\frac{c_{+, \infty} e^{-zF\Phi/RT}}{1 + \tilde{c}_{+, \infty} (e^{-zF\Phi/RT} - 1)} - \frac{c_{-, \infty} e^{zF\Phi/RT}}{1 + \tilde{c}_{-, \infty} (e^{zF\Phi/RT} - 1)} - z\rho \right], \quad \Phi_L = \Phi_{dc} \\ \Phi_R &= 0 \quad \nabla \Phi_R = 0 \end{aligned} \quad (11)$$

This expression can be resolved and gives the electric field E ³⁸

$$E = \left(\frac{2RTc_{\max}}{\epsilon_0 \epsilon_i} \{ \ln[1 + \tilde{c}_{+, \infty} (e^{-zF\Phi/RT} - 1)] + \ln[1 + \tilde{c}_{-, \infty} (e^{zF\Phi/RT} - 1)] \} + \frac{2F\Phi\tilde{\rho}c_{\max}}{\epsilon_0 \epsilon_i} \right)^{1/2} \quad (12)$$

In eq 12, $c_{\max}$ is the maximum concentration of charges in the material. The analytical space charge layer capacitance C_s is defined from the surface charge density Q and the electrostatic potential Φ :

$$C_s = -\frac{\partial Q}{\partial \Phi} = -\frac{\partial(-A\epsilon_0 \epsilon_i E)}{\partial \Phi} = A\epsilon_0 \epsilon_i \frac{\partial E}{\partial \Phi} \quad (13)$$

Finally, we derive the analytical space charge layer capacitance C_s of the solid-state electrolyte, which is expressed by

$$\begin{aligned} C_s &= A \left(Fc_{\max} \left[\frac{-z\tilde{c}_{+, \infty} e^{-zF\Phi/RT}}{1 + \tilde{c}_{+, \infty} (e^{-zF\Phi/RT} - 1)} + \frac{z\tilde{c}_{-, \infty} e^{zF\Phi/RT}}{1 + \tilde{c}_{-, \infty} (e^{zF\Phi/RT} - 1)} \right] + F\tilde{\rho}c_{\max} \right) \\ &\quad / \left(\frac{2RTc_{\max}}{\epsilon_0 \epsilon_i} \{ \ln[1 + \tilde{c}_{+, \infty} (e^{-zF\Phi/RT} - 1)] + \ln[1 + \tilde{c}_{-, \infty} (e^{zF\Phi/RT} - 1)] \} + \frac{2F\Phi\tilde{\rho}c_{\max}}{\epsilon_0 \epsilon_i} \right)^{1/2} \end{aligned} \quad (14)$$

To verify the analytical solution, we compare them with the numerical results of the MPNP model. Different initial concentrations are considered in the solid-state electrolyte, in which mobile lithium ions are $\tilde{c}_+ = 0.5$ and electrons are $\tilde{c}_- = 0.1$ and 0.5 . The boundary conditions fulfill our assumptions and are expressed by $\tilde{J}_{(+, L)} = \tilde{J}_{(-, L)} = 0$ and $\tilde{c}_{(+, R)} = 0.5$, $\tilde{c}_{(-, R)} = 0.1, 0.5$. The metal/electrolyte/metal cell operates with the potentiostatic condition and the given values are $\tilde{\Phi}_L = 0.01$ and $\tilde{\Phi}_R = 0$. Other parameters applied in the simulation are given in the Table 1.

Figure 2 plots the equilibrium state with two different initial concentrations in the solid-state electrolyte. In the legend, "Numerical" represents the calculated results from the MPNP model, and "Analytical" indicates the derived results, which are given by eqs 10 and 12. Note that the analytical and numerical results coincide well with each other as shown in Figure 2 and indicate that the analytical solutions are correct. Moreover, we followed the method presented in Wang's work²⁰ and show the specific capacitance as a function of frequencies in Figure 2c,f. In the citation, the specific capacitance is calculated by $C = -1/(\omega Z_{\text{im}})$ and the value at the lowest frequency marked by the arrow in Figure 2c,f should be equal to the space charge layer capacitance C_s . The imaginary impedance Z_{im} results are taken from the following impedance subsection. According to eq 14, the estimated space charge layer capacitance is given by $C_s = 6.02 \times 10^{-6}, 7.29 \times 10^{-6}$ F, when the electron initial dimensionless concentration $\tilde{c}_- = 0.1$ and 0.5 , respectively. It is evident that the numerical value equals to the analytical space charge layer capacitance C_s predicted by eq 14 for different concentrations. This confirms that our analytical space charge layer capacitance based on the MPNP model is accurate and can apply in any situations, e.g., immobile electrons ($D_- = 0$) or unequal mobile species concentrations ($\tilde{c}_- \neq \tilde{c}_+$).

In some investigations, the space charge layer capacitance C_s^* is also assumed to be correlated with the Debye length λ_D and is given by eq 15

$$\lambda_D = \sqrt{\frac{RT\epsilon_i\epsilon_0}{F^2 \sum z_i^2 c_i}} \quad C_s^* = A \frac{\epsilon_i \epsilon_0}{\lambda_D} \quad (15)$$

However, we have to notice that eq 15 holds merely when the interfacial charge density is zero, i.e., $\tilde{c}_+ - \tilde{c}_- = 0$.⁵² As clearly shown in Figure 2a,d, the charge separation happens in the space charge layer and eq 15 is not suitable to calculate the interfacial capacitance anymore. Moreover, the comprehensive comparison between these two capacitances C_s and C_s^* are shown in the following subsection.

3.2. Impedance Analysis of Ideally Blocking Solid-State Electrolytes. We introduce the analytical space charge layer capacitance C_s based on the advanced MPNP model in Section 3.1. However, the space charge layer resistance R_s due to the charge accumulation or depletion²⁶ and the space charge layer polarization resistance R_s^p are not discussed yet. Thus, a particular discussion about the space charge layer impedance Z_i is presented in this subsection. The corresponding equivalent circuit model for the metal/electrolyte/metal cell and the material properties quantification for elements are given in Figure 1d. In order to reduce the influence factors and simplify the consideration, the concentration boundary conditions of the solid-state electrolyte are given by $J_{(+,L)} = J_{(-,L)} = 0$ on the left-hand side and $\tilde{c}_{(+,R)} = \tilde{c}_0$, $\tilde{c}_{(-,R)} = \tilde{c}_0$ on the right-hand side, respectively. No charge accumulation and the space charge layer formation is assumed at the right interface; thus, $Z_{(+,R)} = Z_{(-,R)} = 0$ as shown in Figure 1d. The conductivity σ_i and the resistance R_i of the species i in the solid-state electrolyte are expressed by⁴⁸

$$\sigma_i = \frac{z_i^2 F^2 D_i c_i}{RT} \quad R_i = \int_0^{L_c} \frac{dx}{A \sigma_i} = \frac{RTL_e}{z_i^2 F^2 A D_i c_i} \quad (16)$$

Moreover, the capacitance of the solid-state electrolyte is given by

$$C_e = A \frac{\epsilon_0 \epsilon_i}{L_e} \quad (17)$$

In a previous study,¹⁷ the interfacial impedance $Z_{(+,L)}$ and $Z_{(-,L)}$ consist of the space charge layer capacitance $C_{s,i}$ and the space charge layer resistance $R_{s,i}$ which are at the equilibrium state. Both of them are caused by the charge accumulation or depletion in the space charge layer²⁶ and have a significant influence on the total interfacial impedance. The total space charge layer capacitance C_s is given by eq 14 and the specific capacitance $C_{s,i}$ of species i is proportional to the charge density and is depicted by

$$C_{s,+} = \frac{\tilde{c}_+}{\tilde{c}_+ + \tilde{c}_-} C_s \quad C_{s,-} = \frac{\tilde{c}_-}{\tilde{c}_+ + \tilde{c}_-} C_s \quad (18)$$

According to the work by Jamnik and Maier,²² the conductivity σ_i in the space charge layer is taken as an independent parameter. The space charge layer resistance $R_{s,i}$ is merely caused by deviation from the bulk contribution and the result for this term is

$$R_{s,i} = \frac{RT}{z_i^2 F^2 A D_i} \int_0^{\lambda_s} \frac{dx}{\tilde{c}_i} \quad (19)$$

where λ_s is the length of the space charge layer and D_i is the bulk diffusivity. Recently, Klerk⁹ pointed out that the diffusivity in the space charge layer is strongly dependent on the concentration and differs from the bulk diffusivity. The diffusivity is given by $D_i = D_i^* \tilde{c}_i (1 - \tilde{c}_i)$ for the solid-state electrolyte and reaches the maximum value when the half regular lattices sites are occupied. Note that these two impedances are calculated when the electrochemical cell at equilibrium state, i.e., $\omega = 0$. Unfortunately, the charge density in the space charge layer depends on the frequencies during the EIS measurement. Thus, the capacitance C_s and resistance R_s are functions of frequencies³⁴ and cannot explain the polarization and perturbation effect with stable values as shown in the literature.^{33,50,51} In order to solve this issue, we introduce the space charge layer polarization impedance R_s^p for the first time. The polarization resistance is related to the electrolyte resistance R_i and the perturbation frequency ω , therefore is given by

$$R_s^p = \frac{R_+ R_-}{R_+ + R_-} \frac{1}{\omega^\alpha} \quad (20)$$

in which α is a coefficient representing the interface blocking level. More exactly, $\alpha = 1$ when the solid-state electrolyte is the ideal blocking for the lithium ions. The total space charge layer impedance is the summary of these three parts:

$$Z_{(+,L)} = R_{s,+} + 1/(j\omega C_{s,+}) + R_s^p \quad Z_{(-,L)} = R_{s,-} + 1/(j\omega C_{s,-}) + R_s^p \quad (21)$$

In the transmission line shown in Figure 1d, C_i^δ is called the chemical capacitance and is determined by the chemical potential μ_i of species i .²⁵ In the MPNP model, the vacancy effect in the solid-state electrolyte is considered and the chemical potential is given by $\mu_i = \mu_i^\ominus + RT \ln \tilde{c}_i - RT \ln(1 - \tilde{c}_i)$. Thus, the chemical capacitance C_i^δ of species i is

$$C_i^\delta = (Fz)^2 A L_e \left(\frac{\partial \mu_i}{\partial c_i} \right)^{-1} \quad (22)$$

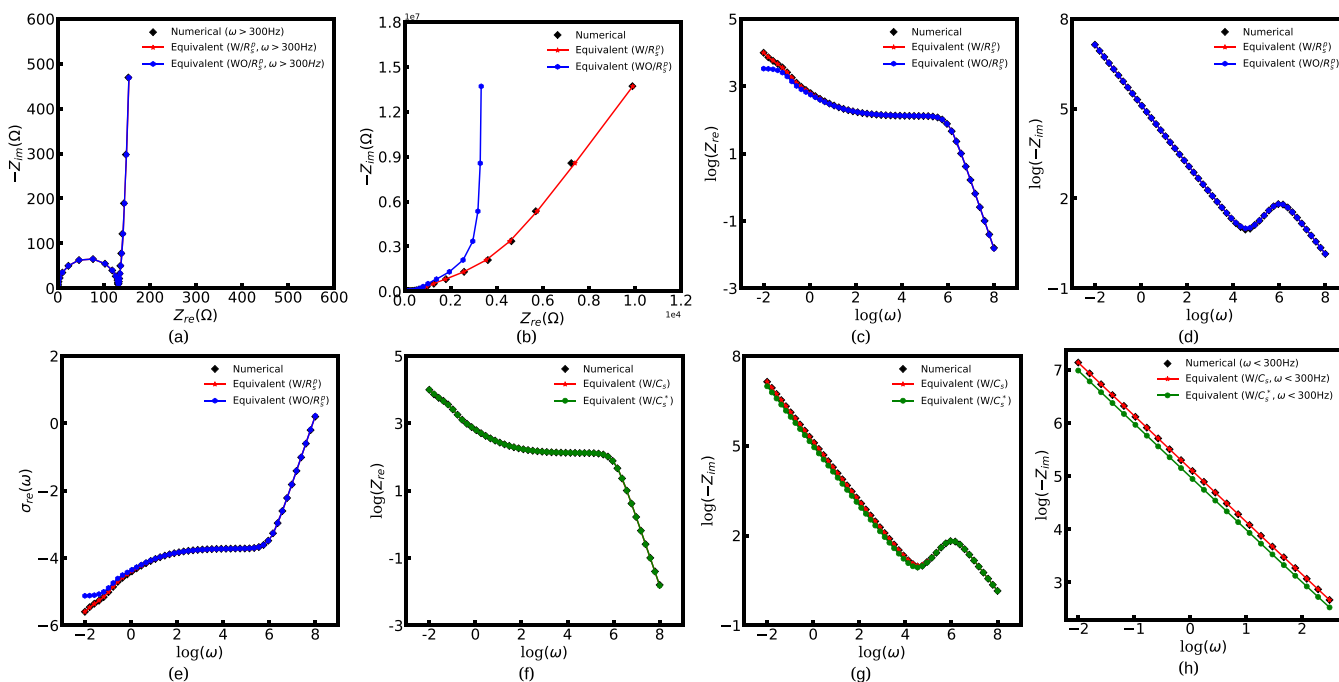


Figure 3. Nyquist plots of the solid-state electrolyte with and without the space charge layer polarization resistance: (a) $300 < \omega$, (b) $10^{-2} < \omega < 10^8$, (c, d) the corresponding real and imaginary Bode plots, and (e) the ionic conductivities. (f) The real Bode plot with different space charge layer capacitances, eqs 14 and 15 are plotted. The imaginary Bode plots: (g) $10^{-2} < \omega < 10^8$ Hz and (h) $\omega < 300$ Hz.

The electrolyte holds local electroneutrality in the homogeneous region; thus, ionic and electronic chemical capacitors are connected in series.²² The total chemical capacitance C^δ is the summary of two species and is depicted by

$$C^\delta = \frac{1}{\sum (1/C_i^\delta)} = \frac{c_{\max}(Fz)^2 AL}{RT} \left[\frac{1}{\tilde{c}_+(1 - \tilde{c}_+)} + \frac{1}{\tilde{c}_-(1 - \tilde{c}_-)} \right] \quad (23)$$

According to the boundary conditions and all the above impedances, the transmission impedance Z_w in the red rectangular in Figure 1d and the total impedance of the solid-state electrolyte are presented by

$$Z_w = (R_+ R_-) / R_w + ([R_+^2 Z_-(L) + R_-^2 Z_+(L)] \tanh[\sqrt{j\omega C^\delta R_w}] + Z_-(L) Z_+(L) R_w \sqrt{j\omega C^\delta R_w}) / (R_w [R_w \tanh[\sqrt{j\omega C^\delta R_w}] + \sqrt{j\omega C^\delta R_w} [Z_+(L) + Z_-(L)]] \quad (24)$$

$$Z = \frac{1}{j\omega C_e + 1/Z_w} \quad (25)$$

and $R_w = R_+ + R_-$ in eq 24.

To verify our equivalent circuit model of the metal/electrolyte/metal cell, the contrasted electrochemical impedance spectroscopy with different diffusivities is presented with the MPNP model. The applied frequency ω of the perturbation potential is from 10^{-2} to 10^8 Hz, and 50 samples of the corresponding current are recorded. The amplitude of the perturbation potential is $\tilde{\Phi}_{ac} = 0.01$ and without the direct signal $\tilde{\Phi}_{dc} = 0$. The initial dimensionless concentrations of mobile lithium ions and electrons in the solid-state electrolyte are $\tilde{c}_+ = \tilde{c}_- = 0.5$ and the diffusivities are $D_+ = 10^{-14}$ and $D_- = 10^{-16} \text{ m}^2 \text{ s}^{-1}$,

respectively. The concentration distribution at equilibrium state is shown in Figure 2, and the thickness of the space charge layer is around $\lambda_s = 1$. Because of the space charge layer length and the slight concentration difference, the diffusivity influence D_i and the space charge layer resistance $R_{s,i}$ given in eq 19 are ignored in our equivalent circuit model, which means $R_{s,i} = 0$.

Figure 3 is the electrochemical impedance spectroscopy of the metal/electrolyte/metal cell and each point depicted in the curve corresponds to the specific frequency. The blue line (Equivalent WO/ R_s^p) represents the space charge layer without the polarization impedance, which is given by $Z_{(i,L)} = 1/(j\omega C_{s,i}) + R_{s,i}$. The red line (Equivalent W/ R_s^p) indicates that the polarization impedance is included in the space charge layer as shown in eq 21. The green line (Equivalent W/ C_s^*) is the result calculated from the commonly applied space charge layer capacitance C_s^* in eq 15. It can be observed that the numerical results of the equivalent circuit model fit the MPNP model well, while differences still exist in some parts.

Figure 3a is the Nyquist plot ($300 \text{ Hz} < \omega$) of the solid-state electrolyte and exhibits a typical curve of the blocking electrolyte, which has a characteristic semicircle at high-frequency zone and a tail at low frequencies. The semicircle represents that the solid-state electrolyte capacitor C_e is in parallel with the electrolyte resistance R_i . The tail is caused by the space charge layer at the metal/electrolyte interface, and the influence is still under discussion.^{9,10} In some investigations, the space charge layer is merely regarded as a pure ideal capacitor³¹ or sometimes the space charge layer resistance R_s is also taken into consideration. However, shown in the literature,^{33,34} the charge density and the space charge layer impedance are frequency-dependent. In order to comprehensively investigate the space charge layer, a polarization resistance R_s^p is included in the proposed equivalent circuit model and exhibits a large influence at low frequencies.

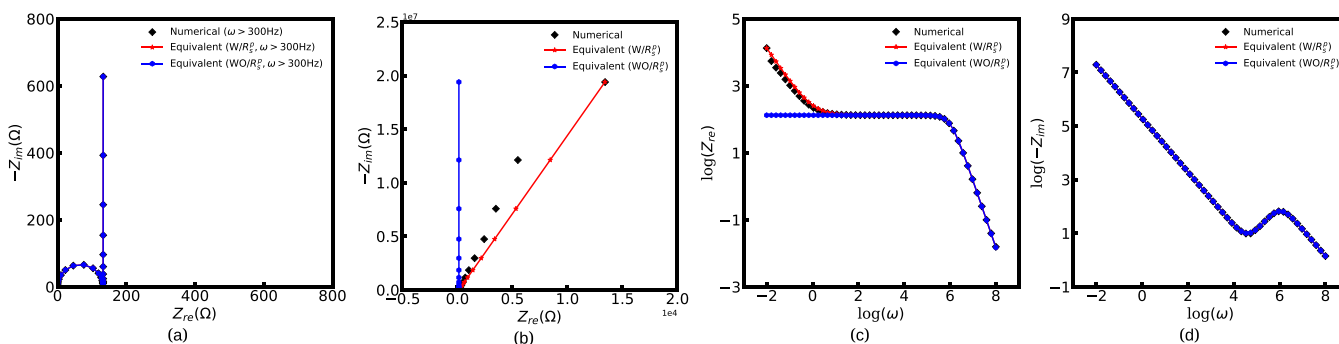


Figure 4. Nyquist plots of the solid-state electrolyte with and without the space charge layer polarization resistance: (a) $300 \text{ Hz} < \omega$, (b) $10^{-2} < \omega < 10^8$, and (c, d) the corresponding real and imaginary Bode plots.

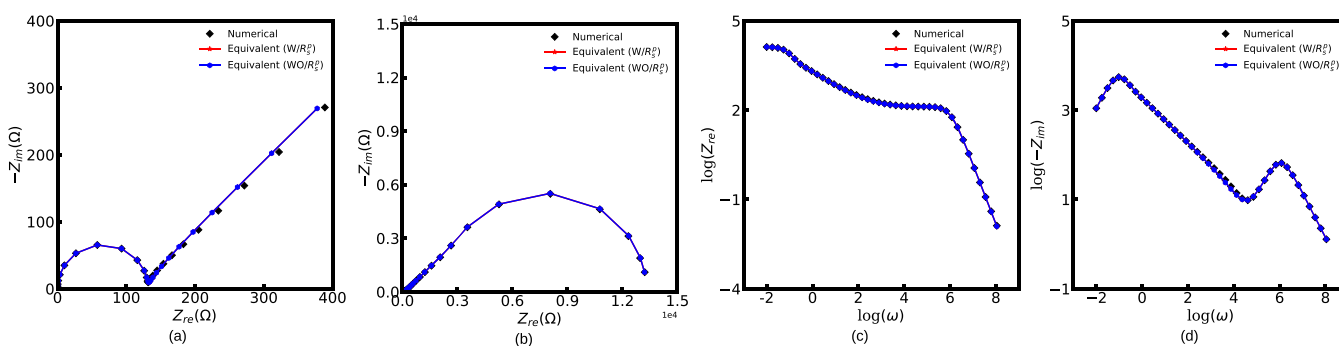


Figure 5. (a, b) Nyquist plots of the solid-state electrolyte with and without the space charge layer polarization resistance and (c, d) the corresponding real and imaginary Bode plots.

Figure 3b plots the overall results ($10^{-2} < \omega < 10^8 \text{ Hz}$) of these two cases. It is evident that the space charge layer without the polarization resistance R_s^p fails to match the numerical results at low frequencies. Figure 3c and Figure 3d are the real and imaginary Bode plots, respectively, and the numerical results (MPNP) are well fitted with our new equivalent circuit model by introducing the polarization resistance. Moreover, it can be noted that the polarization resistance plays an important role at low frequencies. Figure 3e is the ionic conductivity curve of the solid-state electrolyte. This curve is similar to the experimental results of the LiPON thin film as given in the literature^{31,32,50,51} and will be discussed in detail in the following part. To further confirm our numerical results, we follow the method of Yu⁵¹ and calculate the ionic conductivity σ_+ of the solid-state electrolyte. The result is calculated from the real impedance Z_{re} at the frequency ω at which $-Z_{im}$ goes through a local minimum, and this feature is also verified in both of our numerical and equivalent circuit models. Figure 3f and Figure 3h are the Bode plots with different space charge layer capacitances. C_s^* is the extensively applied method²⁰ to estimate the space charge layer capacitance and is shown by eq 15. We can note that the derived space charge layer capacitance C_s exhibits more accurately than C_s^* when fitting the impedance results.

In real materials, the electronic conductivity of the solid-state electrolyte is extremely low and can be assumed as immobile electrons when compared with lithium ions, e.g., $\sigma_+ = 10^{-6}$ and $\sigma_- = 10^{-15} - 10^{-12}$ in LiPON.⁴⁰ Thus, the electrons diffusivity is assumed $D_- = 0$ and the initial concentrations are $\tilde{c}_+ = \tilde{c}_- = 0.5$. Because of the immobile electrons, the electronic line in the transmission line in Figure 1d is ignored. In the corresponding equivalent circuit model, the electrons resistance $R_- = 0$ and the space charge layer capacitance C_s is merely caused by lithium ions and is given by

$$C_{s,+} = C_s$$

$$= A \frac{F c_{\max} \left[\frac{-z \tilde{c}_{+, \infty} e^{-z F \Phi / RT}}{1 + \tilde{c}_{+, \infty} (e^{-z F \Phi / RT} - 1)} \right] + F \tilde{c}_+ c_{\max}}{\sqrt{\frac{2 R T c_{\max}}{\epsilon_0 \epsilon_i} \ln[1 + \tilde{c}_{+, \infty} (e^{-z F \Phi / RT} - 1)]} + \frac{2 F \Phi \tilde{c}_+ c_{\max}}{\epsilon_0 \epsilon_i}} \quad (26)$$

The total space charge layer impedance are $Z_{(+,L)} = R_{s,+} + 1/(j\omega C_{s,+}) + R_s^p$ and $Z_{(-,L)} = 0$, respectively. The polarization capacitance is connected in series with the ionic resistance²⁵ and the transmission line impedance Z_w is

$$Z_w = R_+ + Z_{(+,L)} + \frac{1}{j\omega C^\delta} \quad (27)$$

To better understand the space charge layer polarization resistance, the impedance spectroscopies with immobile electrons in the solid-state electrolyte are shown in Figure 4. When the space charge layer is assumed to be a capacitor, i.e., the polarization resistance R_s^p is ignored, the corresponding equivalent circuit model fails to show the polarization effect as shown in Figure 4c. The reason is that lithium ions deplete in the space charge layer and lead to a drastic potential drop at low frequencies. The depletion layer and the potential are frequency-dependent; unfortunately, the pure capacitor C_s and the space charge layer resistance $R_{s,i}$ are unable to explain this effect. In our equivalent circuit model, the polarization impedance R_s^p is included in the space charge layer and the impedance results fit quite well. The slight difference is caused by the total current deviation. From the above discussion, we can conclude that the pure capacitor assumption is unable to explain the real impedance tail, which is shown in Figure 4c, and the space

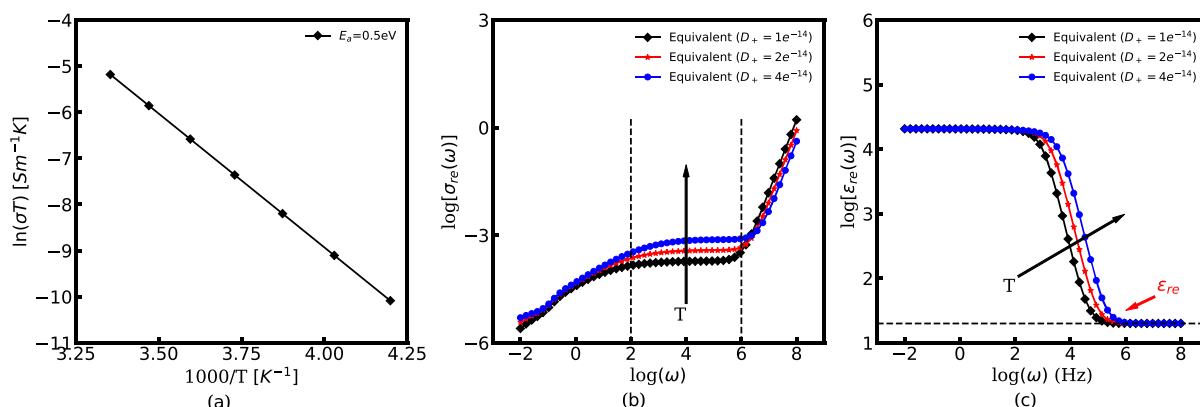


Figure 6. (a) Temperature dependence of the ionic conductivity and (b, c) frequency dependence of the real ionic conductivity and permittivity with different diffusivities.

charge layer polarization resistance R_s^p has to be considered for a perfect fit.

3.3. Impedance Analysis of Non-ideally Blocking Solid-State Electrolytes. In Section 3.2, we discuss an ideally blocking electrolyte for both lithium ions and electrons. Nevertheless, electrons cannot ideally block and may transfer to the blocking electrode, e.g., Au and Pt. To further verify our proposed circuit model and the elements qualifications, a non-blocked solid-state electrolyte for electrons is presented in this section. The initial concentrations of lithium ions and electrons in the solid-state electrolyte are $\tilde{c}_+ = \tilde{c}_- = 0.5$, and other parameters are taken from Table 1. The concentration boundary conditions are given by $J_{(+, L)} = 0$ and $\tilde{c}_- = 0.5$ on the left-hand side and $\tilde{c}_{(+, R)} = \tilde{c}_0$, $\tilde{c}_{(-, R)} = \tilde{c}_0$ on the right-hand side, respectively.

Based on the circuit model shown in Figure 1d, the transmission line impedance Z_w of non-ideally blocking electrolytes and the total impedance lead to

$$Z_w = \frac{R_+ R_-}{R_w} + \frac{R_-^2 Z_{+, (L)}}{R_w^2 + \cosh(\sqrt{j\omega C^\delta R_w}) [R_w Z_{+, (L)} \sqrt{j\omega C^\delta R_w}]} \quad (28)$$

Figure 5a and Figure 5b are the Nyquist plots of the solid-state electrolyte at high frequencies and the overall results. The semicircle in Figure 5a and Figure 5b attributes to the solid-state electrolyte bulk capacitor C_e and resistance R_i and is also observed in Figure 3a. It can be noted that the numerical and analytical impedance curves fit quite well and show the applicability of our circuit model. Nevertheless, with comparison to Figures 3 and 4, the space charge layer effect cannot be observed in Figure 5. This phenomenon attributes to non-blocking electrons, shown in Figure 1d, the total space charge layer impedance consists of ionic and electronic lines, i.e., $Z_{+, (L)}$ and $Z_{-, (L)}$. When electrons are non-blocking in solid-state electrolytes, i.e., $\tilde{c}_{-, (L)} = 0.5$, the electronic line interfacial impedance yields $Z_{-, (L)} = 0$. Therefore, the ionic line is a short circuit under this condition and lithium-ion interfacial impedance $Z_{+, (L)}$ is overlooked at low frequencies.

3.4. Impedance Analysis of Grain Boundaries in Solid-State Electrolytes. In the above sections, we presented a specific discussion with ideally blocking and non-ideally interfaces. However, the grain and grain boundary impedances are also difficult to analyze and puzzle researchers. According to the literature,^{26,53–55} lithium ions will form a space charge layer in the grain boundary region like an interface. Here, we will

utilize our proposed model to deeper the understanding of the grain and grain boundary impedances.

When the grain boundary impedance is much larger than the grain resistance, the grain boundary is similar to an ideally blocking interface. Thus, the total grain impedance can be estimated by eqs 24 and (27), and the boundary capacitance can be observed. As shown in a previous study,⁵⁶ the grain boundary resistance of LATP is about one order of magnitude higher than its grain resistance at room temperature. Therefore, the electrolyte impedance curve shows a vertical tail at low frequencies. When the grain boundary impedance can be ignored, i.e., the non-blocking interface, the polarization effect at low frequencies cannot be observed. The impedance curve at the medium-frequency range is a 45° line, shown in Figure 5b, and this phenomenon is also widely observed in experimental results.^{54,55}

3.5. Temperature Influence. We introduced our MPNP model and the corresponding equivalent circuit model for the metal/electrolyte/metal cell impedance calculation in Sections 3.2 and 3.3. In order to confirm our models, we consider the temperature influence and verify our results by the experimental data published in the literature.³² The conductivity and permittivity are calculated using the following equations:

$$\sigma(\omega) = \frac{L}{A} \left[\frac{1}{Z_{re}(\omega)} + \frac{j}{Z_{im}(\omega)} \right] = \sigma_{re}(\omega) + j\sigma_{im}(\omega)$$

$$\epsilon = \frac{\sigma(\omega)}{j\omega\epsilon_0} = \epsilon_{re}(\omega) - j\epsilon_{im}(\omega) \quad (29)$$

According to the experiments, the ionic conductivity σ_+ follows the Arrhenius behavior and is described by the equation

$$T\sigma_+ = \sigma_0 \exp\left(-\frac{E_a}{k_b T}\right) \quad (30)$$

where σ_0 is the pre-exponential factor, E_a is the hopping activation energy, and k_b is the Boltzmann constant. From eq 30, the ionic conductivity σ_+ increases when the temperature T increases, and the plot is shown in Figure 6a. Because the diffusivity is proportional to the ionic conductivity, different lithium ions diffusivities are discussed in order to simplify the calculation. The initial concentrations in the solid-state electrolyte are $\tilde{c}_+ = \tilde{c}_- = 0.5$; however, with different diffusion coefficients, $D_+ = 10^{-14}, 2 \times 10^{-14}, 4 \times 10^{-14} \text{ m}^2\text{s}^{-1}$ and $D_- = 10^{-16} \text{ m}^2\text{s}^{-1}$.

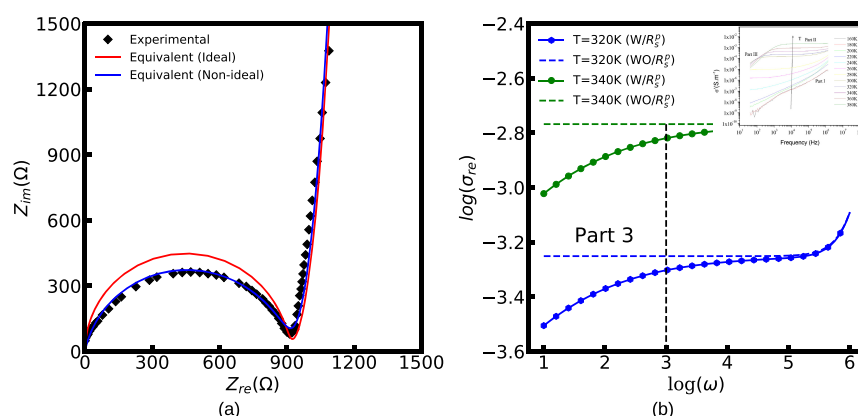


Figure 7. (a) Nyquist plot and (b) conductivity of LiPON for experimental data as presented by ref 32 and the equivalent circuit model. Reproduced with permission from Le Van-Jodin et al.³² Copyright 2013 Elsevier

Figure 6b,c presents the ionic conductivity and permittivity at different temperatures with our equivalent circuit model, which is given in the impedance subsection. In Figure 6b,c, the curves are divided into three parts, i.e., high-frequency ($10^6 < \omega$), medium-frequency ($10^2 < \omega < 10^6$), and low-frequency regions ($\omega < 10^2$), which are similar like the experimental results.^{29,32,57} At high frequencies, the ionic conductivity increases dramatically with the frequencies ω while the permittivity reaches a stable limit. Thus, the material corresponds to a relaxed system without an electric field. Therefore, lithium ions tend to non-randomly hopping and migrate by carrying along their neighbors, in the manner of “jellyfish-type” movement.^{57–59} Moreover, in Figure 6c, the intersection of the dashed line and the permittivity curves corresponds to the equilibrium relative permittivity ϵ_i marked by the arrow as given in Table 1. The intersection value is $\log(\epsilon_{re}) = 1.3$ in Figure 6c, thus the calculated permittivity $\epsilon_{re} = 19.95$ and fulfills the material relative permittivity ϵ_i . This relation can verify that our equivalent circuit model is physical from another perspective. In the medium-frequency region, the ionic conductivity curve shows a stable plateau, which corresponds to the constant value σ_+ in eq 16. Figure 6b shows that the ionic conductivity σ_+ increases with the diffusivity, which also means that the ionic conductivity increases with the temperature. In medium-frequency ranges, lithium ions migrate randomly and are described by eq 30. At low frequencies, the ionic conductivity decreases to a relatively low value due to the polarization effect.³¹ With a low-frequency potential perturbation, electrons accumulate in the space charge layer, then leading to a depletion of lithium ions. The feature causes a large electrostatic potential drop and a drastic fall of the ionic conductivity in this small region. Even though this equivalent circuit model is able to calculate the impedance spectroscopies at different temperatures, the permittivity influence is not included. The solid-state electrolyte permittivity is also changed with the temperature, and this part can be taken into consideration in the future work.

3.6. Comparison with the Experimental Data. We introduced an advanced MPNP model and the corresponding equivalent circuit model for the impedance spectroscopy simulation in the above sections. The calculated impedance spectroscopy shows typical features, which are observed in the experimental results. To further verify our equivalent circuit model, the impedance data, which are taken from Lucie,³² are applied to compare them with our equivalent circuit model. In his work, Pt/LiPON/Pt cell operates with an AC perturbation

and the thickness of solid-state electrolyte LiPON is $L = 1.4 \mu\text{m}$, the active surface area is $A = 1 \times 10^{-5} \text{m}^2$. According to the electrode analysis, the mobile lithium ion concentration is $c_+ = 249.17 \text{mol m}^{-3}$ and the maximum concentration is $c_{\text{max}} = 5.98 \times 10^4 \text{mol m}^{-3}$. The diffusion coefficient of lithium ions is uncertain based on different measurement methods, and the range is from 1.7×10^{-13} to $2 \times 10^{-12} \text{m}^2 \text{s}^{-1}$ at 300 K. Concerning the impedance spectroscopy and other ionic conductivity values of the LiPON thin film,⁶⁰ the lithium ion diffusion coefficient is determined as $D_+ = 1.7 \times 10^{-13} \text{m}^2 \text{s}^{-1}$ in the calculation. Thus, the ionic and electronic conductivities are $\sigma_+ = 1.58 \times 10^{-4}$ and $\sigma_- = 8 \times 10^{-12} \text{S m}^{-1}$, respectively. Moreover, the relative permittivity in his measurement is around $\epsilon_i = 45$ according to the data from Fig. 5 in ref 32. The electronic conductivity is much smaller than the ionic conductivity; thus, the electronic resistance is ignored in the simulation. The relevant equivalent circuit model for the impedance spectroscopy simulation is given by eq 27.

Figure 7a shows the Nyquist impedance plots of the experimental and our equivalent circuit model results. Because the solid-state electrolyte cannot ideally block lithium ions during the experiments, this deviation has to be taken into consideration. As shown in eq 20, α represents the solid-state electrolyte interface blocking for lithium ions and is given as $\alpha = 0.4$ in our calculation. The legend “Equivalent (Ideal)” indicates that the solid-state electrolyte is an ideal capacitor and vice versa. It is evident that the curves fit well at low frequencies but have a slight difference in the high-frequency region. Due to the real material properties, the solid-state electrolyte cannot be regarded as an ideal capacitor. We can see that our ideal equivalent circuit model is not able to show the depressed semi-circle in the experimental curve. In order to tackle this issue, the non-ideal capacitor of the solid-state electrolyte is considered³¹ and is presented by $Z_c = 1/(j\omega C_e)^\beta$. Thus, the depressed semi-circle is clearly shown in our equivalent circuit model with a non-ideal electrolyte. Figure 7b presents the conductivity curves of LiPON thin film layer at different temperatures, and the inserted small figure is results from the literature³² Note that the polarization tail at low frequencies are clear shown in both of our new model and the experimental results. However, this feature cannot be observed in the widely applied circuit model without the polarization impedance R_p^s . This result shows the advantage of our circuit model and clearly verifies the influence of the polarization impedance. We have to notice that all the inputs of the LiPON layer in our model are merely derived from the

citation and may show the deviations, but these results can still show the applicability of our equivalent circuit model.

4. CONCLUSIONS

In this work, we proposed an advanced modified Planck–Nernst–Poisson (MPNP) model and the corresponding equivalent circuit model for the solid-state electrolyte impedance simulation. Within the MPNP framework, the space charge layer formation and the unoccupied regular lattice sites (vacancies) effect in the solid-state electrolyte is considered. In the proposed equivalent circuit model, the elements are quantified and related to the solid-state electrolyte properties. Verified by the experimental impedance results (Pt/LiPON/Pt), this equivalent circuit model is physically meaningful and can calculate the solid-state electrolyte impedance, which includes the space charge layer.

To figure out the high interfacial impedance, the analytical space charge layer capacitance is derived based on the MPNP model. The concentrations of mobile species in the solid-state electrolyte are discussed. Thus, the analytical capacitance is not only limited to the neutrality case. We also discuss the compositions of space charge layer impedance in detail and introduce the polarization resistance in the corresponding equivalent circuit model. The polarization effect is caused by the perturbed charge density in the space charge layer. Numerical results indicate that the polarization resistance has a critical influence on the impedance spectroscopy in the low-frequency region, especially for the electrolyte with immobile electrons. Moreover, the calculated spectroscopies clearly show that the ionic conductivities increase with the temperature and fulfill the Arrhenius equation. To verify our equivalent circuit model further, the relative permittivity is calculated from the spectroscopies and equals to the material property. This study presents a physical equivalent circuit model to calculate the solid-state electrolyte impedance and provides specific explanations about the space charge layer impedance. The material properties quantifications are also studied in order to investigate and optimize the high interfacial impedance of the solid-state electrolyte. Furthermore, we apply this model to analyze the grain and grain boundary inside materials and provide a perspective to qualitatively explain the impedance plots. This model will be used to calculate the full cell impedance, which considers cathode/electrolyte interface reaction in the near future.

Whereas the vacancy effect may be considered to be rather small for the free electrolyte and contact (Pt/LiPON), we expect strong deviations from the simplified capacitor interface model for less ideal material's combination, where the space charge layer capacitance may play a more dominant role.

AUTHOR INFORMATION

Corresponding Authors

René Hausbrand – Surface Science Division, Department of Materials Science, TU Darmstadt, 64287 Darmstadt, Germany; orcid.org/0000-0003-4601-4769; Email: hausbrand@surface.tu-darmstadt.de

Bai-Xiang Xu – Mechanics of Functional Materials Division, Department of Materials Science, TU Darmstadt, 64287 Darmstadt, Germany; Email: xu@mfm.tu-darmstadt.de

Authors

Yao Liu – Mechanics of Functional Materials Division, Department of Materials Science and Surface Science Division,

Department of Materials Science, TU Darmstadt, 64287 Darmstadt, Germany; orcid.org/0000-0003-0363-8803

Yang Bai – Mechanics of Functional Materials Division, Department of Materials Science, TU Darmstadt, 64287 Darmstadt, Germany

Wolfram Jaegermann – Surface Science Division, Department of Materials Science, TU Darmstadt, 64287 Darmstadt, Germany

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsami.0c22986>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors gratefully acknowledge Wen-bei Yu for the discussion and the financial support from the China Scholarship Council (CSC). The calculations for this study have been conducted on the “Lichtenberg” high-performance computer of the TU Darmstadt, which we gratefully acknowledge.

REFERENCES

- (1) Liu, Y.; Zhu, Y.; Cui, Y. Challenges and Opportunities towards Fast-Charging Battery Materials. *Nat. Energy* **2019**, *4*, 540–550.
- (2) Yamada, Y.; Wang, J.; Ko, S.; Watanabe, E.; Yamada, A. Advances and Issues in Developing Salt-Concentrated Battery Electrolytes. *Nat. Energy* **2019**, *4*, 269–280.
- (3) Choi, J. W.; Aurbach, D. Promise and Reality of Post-Lithium-Ion Batteries with High Energy Densities. *Nat. Rev. Mater.* **2016**, *1*, 16013.
- (4) Goodenough, J. B.; Park, K.-S. The Li-Ion Rechargeable Battery: A Perspective. *J. Am. Chem. Soc.* **2013**, *135*, 1167–1176.
- (5) Song, J.; Jacke, S.; Cherkashinin, G.; Schmid, S.; Dong, Q.; Hausbrand, R.; Jaegermann, W. Valence Band Offsets of LiPON/LiCoO₂ Hetero-Interfaces Determined by X-Ray Photoelectron Spectroscopy. *Electrochem. Solid-State Lett.* **2011**, *14*, A189–A191.
- (6) Manthiram, A.; Yu, X.; Wang, S. Lithium Battery Chemistries Enabled by Solid-State Electrolytes. *Nat. Rev. Mater.* **2017**, *2*, 16103.
- (7) Xia, S.; Wu, X.; Zhang, Z.; Cui, Y.; Liu, W. Practical Challenges and Future Perspectives of All-Solid-State Lithium-Metal Batteries. *Chem* **2019**, *5*, 753–785.
- (8) Han, X.; Gong, Y.; Fu, K. K.; He, X.; Hitz, G. T.; Dai, J.; Pearse, A.; Liu, B.; Wang, H.; Rubloff, G.; Mo, Y.; Thangadurai, T.; Wachsman, E. D.; Hu, L. Negating Interfacial Impedance in Garnet-Based Solid-State Li Metal Batteries. *Nat. Mater.* **2017**, *16*, 572–579.
- (9) de Klerk, N. J. J.; Wagemaker, M. Space-charge layers in all-solid-state batteries; important or negligible? *ACS Appl. Energy Mater.* **2018**, *1*, 5609–5618.
- (10) Luntz, A. C.; Voss, J.; Reuter, K. Interfacial Challenges in Solid-State Li Ion Batteries. *J. Phys. Chem. Lett.* **2015**, *6*, 4599–4604.
- (11) Xu, L.; Tang, S.; Cheng, Y.; Wang, K.; Liang, J.; Liu, C.; Cao, Y.-C.; Wei, F.; Mai, L. Interfaces in Solid-State Lithium Batteries. *Joule* **2018**, *2*, 1991–2015.
- (12) Put, B.; Vereecken, P. M.; Meersschaut, J.; Sepúlveda, A.; Stesmans, A. Electrical Characterization of Ultrathin RF-Sputtered LiPON Layers for Nanoscale Batteries. *ACS Appl. Mater. Interfaces* **2016**, *8*, 7060–7069.
- (13) Gittleston, F. S.; El Gabaly, F. Non-Faradaic Li⁺ Migration and Chemical Coordination across Solid-State Battery Interfaces. *Nano Lett.* **2017**, *17*, 6974–6982.
- (14) Dierickx, S.; Mundloch, T.; Weber, A.; Ivers-Tiffée, E. Advanced Impedance Model for Double-Layered Solid Oxide Fuel Cell Cermet Anodes. *J. Power Sources* **2019**, *415*, 69–82.
- (15) MacDonald, J. R. *Impedance Spectroscopy: Emphasizing Solid Materials and Systems*; Wiley-Interscience: 1987.
- (16) Bisquert, J. Theory of the Impedance of Electron Diffusion and Recombination in a Thin Layer. *J. Phys. Chem. B* **2002**, *106*, 325–333.

- (17) Mei, B.-A.; Munteshari, O.; Lau, J.; Dunn, B.; Pilon, L. Physical Interpretations of Nyquist Plots for EDLC Electrodes and Devices. *J. Phys. Chem. C* **2018**, *122*, 194–206.
- (18) Buck, R. P.; Mundt, C. Origins of finite transmission lines for exact representations of transport by the Nernst–Planck equations for each charge carrier. *Electrochim. Acta* **1999**, *44*, 1999–2018.
- (19) Wang, H.; Thiele, A.; Pilon, L. Simulations of Cyclic Voltammetry for Electric Double Layers in Asymmetric Electrolytes: A Generalized Modified Poisson–Nernst–Planck Model. *J. Phys. Chem. C* **2013**, *117*, 18286–18297.
- (20) Wang, H.; Pilon, L. Intrinsic Limitations of Impedance Measurements in Determining Electric Double Layer Capacitances. *Electrochim. Acta* **2012**, *63*, 55–63.
- (21) Becker-Steinberger, K.; Funken, S.; Landstorfer, M.; Urban, K. A Mathematical Model for All Solid-State Lithium-Ion Batteries. *ECS Trans.* **2010**, *25*, 285–296.
- (22) Jamnik, J.; Maier, J. Generalised Equivalent Circuits for Mass and Charge Transport: Chemical Capacitance and its Implications. *Phys. Chem. Chem. Phys.* **2001**, *3*, 1668–1678.
- (23) Jamnik, J. Impedance Spectroscopy of Mixed Conductors with Semi-Blocking Boundaries. *Solid State Ionics* **2003**, *157*, 19–28.
- (24) Hamann, C. H.; Hamnett, A.; Vielstich, W. *Electrochemistry*; Wiley-VCH: 2007.
- (25) Maier, J. *Physical Chemistry of Ionic Materials: Ions and Electrons in Solids*; John Wiley & Sons: 2004; DOI: 10.1002/0470020229.
- (26) Maier, J. On the Conductivity of Polycrystalline Materials. *Bunsen-Ges. Phys. Chem., Ber.* **1986**, *90*, 26–33.
- (27) Liu, Y.; Ma, Y.-B.; Jaegermann, W.; Hausbrand, R.; Xu, B.-X. Interface Equilibrium Modeling of All-Solid-State Lithium-Ion Thin Film Batteries. *J. Power Sources* **2020**, *454*, 227892.
- (28) Wang, H.; Pilon, L. Accurate Simulations of Electric Double Layer Capacitance of Ultramicroelectrodes. *J. Phys. Chem. C* **2011**, *115*, 16711–16719.
- (29) Landstorfer, M.; Funken, S.; Jacob, T. An Advanced Model Framework for Solid Electrolyte Intercalation Batteries. *Phys. Chem. Chem. Phys.* **2011**, *13*, 12817–12825.
- (30) Stegmaier, S.; Voss, J.; Reuter, K.; Luntz, A. C. Li⁺ Defects in a Solid-State Li Ion Battery: Theoretical Insights with a Li₃OCl Electrolyte. *Chem. Mater.* **2017**, *29*, 4330–4340.
- (31) Larfaillou, S.; Guy-Bouysou, D.; Le Cras, F.; Franger, S. Comprehensive Characterization of All-Solid-State Thin Films Commercial Microbatteries by Electrochemical Impedance Spectroscopy. *J. Power Sources* **2016**, *319*, 139–146.
- (32) Le Van-Jodin, L.; Ducroquet, F.; Sabary, F.; Chevalier, I. Dielectric Properties, Conductivity and Li⁺ Ion Motion in LiPON Thin Films. *Solid State Ionics* **2013**, *253*, 151–156.
- (33) Jamnik, J.; Pejovnik, S.; Maier, J. A New Approach for the Computation of the Frequency Response of Space Charge-Containing Interfaces. *Electrochim. Acta* **1993**, *38*, 1975–1978.
- (34) MacDonald, J. R. Theory of Space-Charge Polarization and Electro-Discharge Effects. *J. Chem. Phys.* **1973**, *58*, 4982–5001.
- (35) Braun, S.; Yada, C.; Latz, A. Thermodynamically Consistent Model for Space-Charge-Layer Formation in a Solid Electrolyte. *J. Phys. Chem. C* **2015**, *119*, 22281–22288.
- (36) Fingerle, M.; Buchheit, R.; Siculo, S.; Albe, K.; Hausbrand, R. Reaction and Space Charge Layer Formation at the LiCoO₂–LiPON Interface: Insights on Defect Formation and Ion Energy Level Alignment by a Combined Surface Science–Simulation Approach. *Chem. Mater.* **2017**, *29*, 7675–7685.
- (37) Haruyama, J.; Sodeyama, K.; Han, L.; Takada, K.; Tateyama, Y. Space-Charge Layer Effect at Interface between Oxide Cathode and Sulfide Electrolyte in All-Solid-State Lithium-Ion Battery. *Chem. Mater.* **2014**, *26*, 4248–4255.
- (38) Newman, J.; Thomas-Alyea, K. E. *Electrochemical Systems*; John Wiley & Sons: 2012.
- (39) Bates, J. B.; Dudney, N. J.; Gruzalski, G. R.; Zuhr, R. A.; Choudhury, A.; Luck, C. F.; Robertson, J. D. Fabrication and Characterization of Amorphous Lithium Electrolyte Thin Films and Rechargeable Thin-Film Batteries. *J. Power Sources* **1993**, *43*, 103–110.
- (40) Han, F.; Westover, A. S.; Yue, J.; Fan, X.; Wang, F.; Chi, M.; Leonard, D. N.; Dudney, N. J.; Wang, H.; Wang, C. High Electronic Conductivity as the Origin of Lithium Dendrite Formation within Solid Electrolytes. *Nat. Energy* **2019**, *4*, 187–196.
- (41) Lai, W.; Haile, S. M. Impedance Spectroscopy as a Tool for Chemical and Electrochemical Analysis of Mixed Conductors: A Case Study of Ceria. *J. Am. Ceram. Soc.* **2005**, *88*, 2979–2997.
- (42) Jamnik, J.; Maier, J. Treatment of the Impedance of Mixed Conductors Equivalent Circuit Model and Explicit Approximate Solutions. *J. Electrochem. Soc.* **1999**, *146*, 4183–4188.
- (43) Mei, B.-A.; Lau, J.; Lin, T.; Tolbert, S. H.; Dunn, B. S.; Pilon, L. Physical Interpretations of Electrochemical Impedance Spectroscopy of Redox Active Electrodes for Electrical Energy Storage. *J. Phys. Chem. C* **2018**, *122* (43), 24499–24511.
- (44) Song, J.; Bazant, M. Z. Effects of Nanoparticle Geometry and Size Distribution on Diffusion Impedance of Battery Electrodes. *J. Electrochem. Soc.* **2012**, *160*, A15–A24.
- (45) Jamnik, J.; Habermeier, H.-U.; Maier, J. Information of Ionic Boundary Effects by a Novel Penetration Impedance Technique. *Phys. B* **1995**, *204*, 57–64.
- (46) Jamnik, J. Impact of Particle Size on Conductivity and Storage Capacity as Derived from the Core-Space Charge Model. *Solid State Ionics* **2006**, *177*, 2543–2547.
- (47) Meyers, J. P.; Doyle, M.; Darling, R. M.; Newman, J. The Impedance Response of a Porous Electrode Composed of Intercalation Particles. *J. Electrochem. Soc.* **2000**, *147*, 2930–2940.
- (48) Huang, J.; Li, Z.; Ge, H.; Zhang, J. Analytical Solution to the Impedance of Electrode/Electrolyte Interface in Lithium-Ion Batteries. *J. Electrochem. Soc.* **2015**, *162*, A7037–A7048.
- (49) Gaston, D.; Newman, C.; Hansen, G.; Lebrun-Grandié, D. MOOSE: A Parallel Computational Framework for Coupled Systems of Nonlinear Equations. *Nucl. Eng. Des.* **2009**, *239*, 1768–1778.
- (50) Nimisha, C. S.; Rao, K. Y.; Venkatesh, G.; Rao, G. M.; Munichandraiah, N. Sputter Deposited LiPON Thin Films from Powder Target as Electrolyte for Thin Film Battery Applications. *Thin Solid Films* **2011**, *519*, 3401–3406.
- (51) Yu, X.; Bates, J. B.; Jellison, G. E., Jr.; Hart, F. X. A Stable Thin-Film Lithium Electrolyte: Lithium Phosphorus Oxynitride. *J. Electrochem. Soc.* **1997**, *144*, 524.
- (52) Sato, N. *Electrochemistry at Metal and Semiconductor Electrodes*; Elsevier: 1998.
- (53) Goswami, N.; Kant, R. Theory for impedance response of grain and grain boundary in solid state electrolyte. *J. Electroanal. Chem.* **2019**, *835*, 227–238.
- (54) Liu, Y.-H.; Chen, K. T.; Liu, B. H. Fabrication of composite probe electrode used for localized impedance analysis of solid-state electrolyte LATP. *Solid State Ionics* **2019**, *336*, 11–18.
- (55) Zhu, Y.; Wu, S.; Pan, Y.; Zhang, X.; Yan, Z.; Xiang, Y. Reduced Energy Barrier for Li⁺ Transport Across Grain Boundaries with Amorphous Domains in LLZO Thin Films. *Nanoscale Res. Lett.* **2020**, *15*, 153.
- (56) Mariappan, C. R.; Gellert, M.; Yada, C.; Rosciano, F.; Roling, B. Grain boundary resistance of fast lithium ion conductors: comparison between a lithium-ion conductive Li–Al–Ti–P–O-type glass ceramic and a Li_{1.5}Al_{0.5}Ge_{1.5}P₃O₁₂ ceramic. *Electrochem. Commun.* **2012**, *14*, 25–28.
- (57) Mariappan, C. R.; Govindaraj, G. Conductivity and ion dynamic studies in the Na_{4.7+x}Ti_{1.3-x}(PO₄)_{3.3-x} (0 ≤ x ≤ 0.6) NASICON material. *Solid State Ionics* **2005**, *176*, 1311–1318.
- (58) Dyre, J. C.; Maass, P.; Roling, B.; Sidebottom, D. L. Fundamental questions relating to ion conduction in disordered solids. *Rep. Prog. Phys.* **2009**, *72*, No. 046501.
- (59) Bérardan, D.; Franger, S.; Meena, A. K.; Dragoe, N. Room temperature lithium superionic conductivity in high entropy oxides. *J. Mater. Chem. A* **2016**, *4*, 9536–9541.
- (60) Choi, C. H.; Cho, W. I.; Cho, B. W.; Kim, H. S.; Yoon, Y. S.; Tak, Y. S. Radio-Frequency Magnetron Sputtering Power Effect on the Ionic Conductivities of LiPON Films. *Electrochem. Solid-State Lett.* **2002**, *5*, A14–A17.